CONVERSATIONAL GRAPH EXTRACTION FROM UNSTRUCTURED DATA

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Sprint 03

THE CORE PROBLEM

To building a Knowledge Graph from scratch requires two things:

- **schema:** the set of the entity type and relation you want to extract
- **instances**

The schema is the expensive part (domain experts spend a lot of time designing it by hand) and our purpose is to replace the complex process with a conversation, where the user sits in front of a chat, the LLM propose a initial schema from a small sample of docs and the user refines it through natural language.

That's the active system. But the real question is: **how do we know if it worked?**

WHY EVALUATION IS HARD HERE?

The standard evaluation is to compare againist a gold standard, but because the whole point of the projec is to work on **arbitrary domain**, we need a **ground-truth-free evaluation framework.**

Every metric we compute has to come from things **we can observe**: the structure of the conversation, the structure of the extracted graph, and the internal consistency of the schema.

This is actually a well-recognized challenge in the literature, and the papers all deal with it from different angles.

RELATED WORK

- **LLMs4SchemaDiscovery (2025)**: This work looks specifically at using LLMs to induce schemas directly from text. Their key insight is that LLMs, when prompted correctly, can propose ontology structures that are coherent and domain-appropriate. What they don't address is what happens when the proposed schema is too generic or too specific for the actual data which is exactly what our HITL is designed to fix.

- **CLARE: Context-Aware Interactive KG Construction (2025)**: CLARE is the closest work to our system architecturally. It frames KG construction as a dialogue, where the system proposes the user accepts, modifies, or rejects them. What's relevant for us is how they think about the quality of the interaction itself: not just the final graph, but how efficiently the system and user reach agreement.

RELATED WORK

- **Prompt-guided LLM Agents for Ontology Learning (2025):** This paper uses prompted agents that iteratively refine an ontology over multiple rounds, similar to how our HITL works. Their evaluation focuses on structural properties of the final ontology: coverage, redundancy, internal consistency.

- **Benchmarking LLMs for Knowledge Graph Validation (2026)**: This is the most most technically relevant work. This paper benchmarks LLMs on the task of catching schema violations: entities with wrong types, relations with violated domain-range constraints. This directly motivates our **GIV mechanism, Guided Iterative Verification,** where if the LLM's extraction violates the frozen schema, we feed the errors back and ask for a correction.

EVALUATION ARCHITECTURE

Schema quality

structural properties
of the final schema

HITL Convergence

quality of the human-
machine interaction

Generalization

does the schema transfer to
unseen data?

Downstream Utility

does the graph actually
help answer questions?

MAIN SYSTEM PIPELINE

1. The LLM reads the **discovery documents** (10% of the total documents) and proposes an initial schema. This is version 0: **the zero-shot baseline**
2. The user sees the proposed schema in the chat and starts refining it. When is satisfied, they **freeze** the schema.
3. The frozen schema is applied to the **90% validation set**. For each document, extract_document() asks the LLM to populate the schema with instances. If the output violates the schema constraints, the **GIV** (Guided Iterative Verification) repair loop operates.

BLOCK A — SCHEMA QUALITY

Block A measures properties of the final frozen schema against the extracted data.

- **A1. Schema Utilization Rate:**

$$SUR = \frac{\textit{classes with at least one extracted instance}}{\textit{total schema classes}}$$

If you have a schema with 10 entity classes but only 6 of them ever appear in the extracted graph, that's SUR = 0.6. A low SUR means the schema was over-specified and the LLM invented classes during discovery that don't actually appear in the data. This is the signal that the HITL session should have caught and removed those classes.

BLOCK A — SCHEMA QUALITY

Block A measures properties of the final frozen schema against the extracted data.

- **A2. Schema Consistency Rate:**
  - Measures how often the LLM, during batch extraction, violates the frozen schema

$$SCR = 1 - \left(\frac{validation\ errors}{total\ extracted\ triples}\right)$$

- **A3. Orphan Node Rate:**
  - An orphan node is an entity that was extracted but never connected to anything.

$$ONR = \frac{nodes\ with\ degree\ 0}{total\ nodes}$$

BLOCK B - HUMAN IN THE LOOP CONVERGENCE

- **B1. Schema Edit Distance:**
  - At each turn of the conversation, the LLM, in agreement with the user, proposes a set of atomic schema edits: add a class, rename a relation, merge two classes. We compute a weighted sum of those edits.
  - What we hope to see is a decreasing curve: which means that the user and the LLM are coming to a conclusion

BLOCK C — GENERALIZATION

This part is the critical test of whether the schema generalizes beyond the 10% of documents used during discovery.

- **C1. Unmapped Instance Rate:**

$$UIR = \frac{entities\ the\ LLM\ couldn't\ assign\ to\ any\ schema\ class}{total\ extracted\ entities}$$

  - During extraction, when the LLM encounters a mention it can't fit into any of the defined classes, it puts it in an "unmapped" bucket rather than inventing a new class. UIR measures how full that bucket gets.

- **C2. Schema Drift Rate:**

$$SDR = \frac{documents\ where\ the\ LLM\ proposed\ a\ schema\ change}{total\ documents\ processed}$$

BLOCK D — DOWNSTREAM UTILITY

Block D is the most practically meaningful test: does having a knowledge graph actually help answer questions better than just searching the raw text?

1. For each test document, we generate one factual question from its text using a dedicated LLM prompt.
2. We pass that question to each retrieval system and collect an answer.
3. We judge each answer with an LLM-as-judge that receives the question, the answer, and the first 1000 characters of the source document as evidence: it returns YES, PARTIAL, or NO with a brief justification.
4. Finally, we aggregate: accuracy is computed as (YES + 0.5 × PARTIAL) / total, and the key figure is Δ = Accuracy(HITL) - Accuracy(Plain-RAG). If $\Delta > 0$, the graph adds value.

BASELINES

The idea is to compare against these two baselines:

- **Zero-Shot GIV:** the same system, same extraction pipeline, same GIV repair loop, but using the schema produced by the LLM without any HITL refinement. This isolates the contribution of the human-in-the-loop.
- **Plain-RAG:** for each question, it tokenizes the question, removes stopwords, and computes a Jaccard-like overlap score between the question's tokens and every document in the corpus. It returns the three highest-scoring documents and passes their first 1200 characters to the LLM as context.

DATASETS

The idea is to use two datasets: **Reddit AITA posts** and **Wikipedia Historical Events**. The choice is intentional: they have very different structural properties.

Wikipedia Historical Events has structured domain with standardized vocabulary.
AITA is unstructured social narrative: much more subjective, much more ambiguous.

WHAT'S BUILT, WHAT'S NEXT

The timeline:

- **This week**: defined Block D evaluation
- **Next week**: write up results, comparison tables, plots
- **Week after**: finalize the report and presentation

The infrastructure is in place. The backend logs every extraction result, every HITL turn, every LLM call.
All metrics for Blocks A, B, and C are computable automatically from these logs. Block D is a standalone script.